day3-assignment

Day 3 Assignment: Answers

Correct answers to the 15 Day Three assignment questions on Hive and Pig, each with a one-line reason.

Q1. Schema-on-read means the data has no schema.

False. The schema exists in the metastore; Hive applies it when the data is read, not when it is loaded.

Q2. Why are columnar formats like Parquet/ORC preferred for analytical queries?

b. They enable column projection and better compression. A query reads only the columns it needs, and values from one column compress well together.

Q3. Partitioning and bucketing are the same thing.

False. Partitioning splits data into directories by value; bucketing hashes rows into a fixed number of files.

Q4. Bucketing in Hive writes rows into a fixed number of files based on:

c. Hash of a column. The same value always lands in the same bucket file.

Q5. Which UDF type produces many output rows per input row (e.g. splitting a string)?

d. UDTF. One row in, many rows out, as with `explode()`.

Q6. Match each concept to its description.

Concept	Description
Pig Latin	Dataflow scripting language
UDTF	Produces many output rows per input row
Metastore	Central catalog of table and partition definitions
Managed table	Hive owns and drops the underlying data files

Managed versus external comes down to who owns the files.

Q7. What is the role of a SerDe in Hive?

d. Converts between byte streams and rows. SerDe is short for serializer and deserializer.

Q8. What does schema-on-read mean in Hive?

c. Schema is applied when data is read. Traditional databases check the schema on write instead.

Q9. Which type of database is most commonly used as the backend for the Hive metastore?

A relational database (for example MySQL or PostgreSQL). The built-in Derby supports only one user at a time, so it is used for testing.

Q10. Hive partitioning primarily enables:

b. Partition pruning (skip unrelated directories). It only helps when the query filters on the partition column.

Q11. Which statement about Hive's execution engine is correct?

a. Hive is a query layer that can use MapReduce, Tez, or Spark. The engine is set with `hive.execution.engine` (Spark support was removed in Hive 4, so check your version).

Q12. Which Pig complex type is an unordered collection of tuples?

a. Bag. A tuple is an ordered set of fields; a bag is an unordered group of tuples.

Q13. Which statement best describes Pig Latin's programming style?

d. Procedural, dataflow-oriented. A Pig script lists the steps; SQL only states the result you want.

Q14. What does the Hive metastore store?

b. Table, column and partition metadata. The data files themselves stay in HDFS.

Q15. What happens when you DROP an external Hive table?

c. Only metadata is removed; files stay. Dropping a managed table would delete the files too.

Quick recall

- Schema-on-read still has a schema; it is applied at query time.
- Columnar formats: read fewer columns, better compression.
- Partitioning uses directories by value; bucketing uses hash into a fixed number of files.
- UDTF: one input row, many output rows.
- SerDe: bytes to rows and rows to bytes.
- Metastore stores metadata in a relational database; Derby is for testing.
- Partition pruning only works when you filter on the partition column.
- Hive is a query layer on MapReduce, Tez or Spark.
- Pig bag: unordered collection of tuples.
- Pig Latin is procedural dataflow.
- DROP external table: metadata gone, files kept.